**Title:** Optimization of Medical Inventory

**Subtitle:** Optimizing Inventory Strategies for Cost Reduction and Efficient Stock Utilization

**Client:** One of the Leading Biopharmaceutical Companies

**Domain:** Pharmaceutical Inventory and Supply Chain Management

**Presenter:** Jay Bourasi

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# **Summary:** Key Objectives, Business Problem And Insights

- **Key Objectives:** The goal of this project was to optimize stock management by reducing excess inventory, minimizing procurement costs, and improving stock utilization for better overall efficiency.
- **Business Problem:** The company faced issues with high stock levels and inefficient procurement practices, resulting in increased storage costs and underutilized stock. This impacted the overall profitability and operational efficiency.
- **Insights:**
- **Stock Utilization**: Only 10.4% of the total procured stock is still available, indicating either high consumption or poor stock management.
- **Procurement Trends**: Most of the stock is procured by Plant A (90.8%), and Electrical Supplies account for nearly 31% of total procurement.
- **Cost Impact**: Inventory costs have decreased by 87.9% from 2020 to 2024, with the majority of costs still concentrated in **Plant A**, which accounts for 88.54% of the total costs.
- **Material Analysis**: High-cost materials like High-Speed Diesel and Wister Rats are major contributors to inventory costs.

# Business Context

- **Problem**: Inefficiencies in raw material procurement and inventory management have resulted in overpaying and surplus unutilized stock.
- **Objective**: Minimize inventory costs while ensuring effective utilization of existing stock.
- **Constraints**:
- Limited storage capacity.
- Diverse categories of storage locations requiring specific handling.
- **Success Criteria**:
- Business: Reduce underutilized stock levels by 20%.
- Economic: Cut storage and management costs for excess raw materials by 25%.

# Data Overview

- ❑ **Material Information**
- ▪ **Material ID**: A unique identifier for each material.
- ▪ **Material Description**: A brief description of the material.
- ▪ **Material Type**: The type or category of the material (e.g., raw, finished goods).
- ▪ **Material Type Description**: A detailed description of the material type.
- ▪ **Product Description**: A description of the product linked to the material.

- ❑ **Plant and Storage Information**
- ▪ **Plant Code**: The code representing a specific plant or facility.
- ▪ **Plant Name**: The name of the plant or facility.
- ▪ **Storage Location**: The specific location where the material is stored.
- ▪ **Storage Location Category**: The type or category of the storage location (e.g., Quality Control, Bois Storage).

- **Movement and Transaction Details**
  - **Movement Type**: The type of material movement, such as receipt or issue.
  - **Movement Type Description**: A detailed description of the material movement type.
  - **Movement Reason**: The reason associated with the material movement.
  - **Movement Indicator**: A flag indicating the movement type (e.g., Bulk Movement).
  - **Movement Indicator Description**: A description of the movement indicator.
  - **Transaction Event Type**: The type of event that triggered the transaction.
  - **Transaction Event Type Description**: A description of the event type, such as the removal of goods from inventory.
- **Document Tracking and Identification**
  - **Material Document Number**: The unique document number for the material transaction.
  - **Material Document Item Number**: The specific item number within the material document.
  - **Material Document Year**: The year associated with the material document.
  - **Reservation Number**: The reserved material's number, if applicable.
  - **Stock Transfer Item Number**: The item number in a stock transfer transaction.

- **Posting and Entry Information**

- **Posting Date**: The date when the transaction was officially recorded.
- **Document Date**: The date when the document was created.
- **Entry Date**: The date the transaction was entered into the system.
- **Time of Entry**: The time the transaction was entered.

- **Quantity and Units**

- **Quantity in Entry Unit**: The amount of material recorded in the entry unit.
- **Unit of Entry**: The unit of measure used for the entry (e.g., kilograms, numbers).
- **Unit of Entry Category**: The category of the unit of measure.
- **Quantity in Order Price Unit**: The quantity in the pricing unit for the order.
- **Order Price Unit**: The unit used for pricing the material.
- **Order Price Unit Category**: The category of the pricing unit.
- **Base Unit of Measure**: The standard unit used to measure the material.
- **Quantity**: The total quantity of material in the transaction.

- **Financial Information**
  - **Amount in Local Currency**: The value of the transaction in the local currency.
  - **Currency**: The type of currency used for the transaction (e.g., INR).
  - **Debit Credit Indicator**: Indicates whether the transaction is a debit or credit.

- **Order and Reference Information**
  - **Order Reference Number**: A reference number linked to the specific order.
  - **Reference Number**: Additional reference information for tracking.
  - **Purchase Order Number**: The unique number associated with the purchase order.
  - **Purchase Order Item Number**: The line item number in the purchase order.

- **Supplier and Vendor Details**
  - **Supplier Code**: The unique identifier for the supplier.
  - **Vendor Code**: The unique identifier for the vendor.

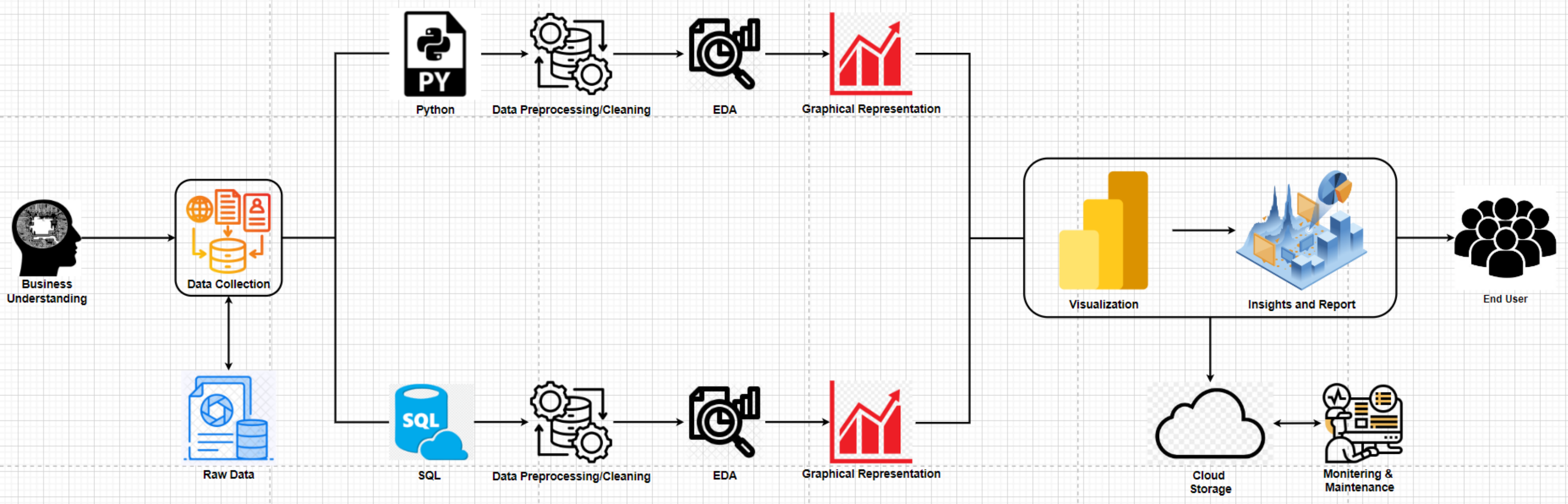
- **Miscellaneous Information**
  - **Company Code**: The code representing the company.
  - **Cost Center**: The cost center associated with the transaction.
  - **User Name**: The name of the user who performed the transaction.
  - **Creation Method**: The method used to create the transaction (manual or automatic).

# Methodology

The project followed a structured approach using various tools:

- **Data Cleaning (Excel):**
  For initial steps include - Removed duplicates, handled null values, and ensured consistency for analysis.
- **EDA & Preprocessing (SQL & Python):**
  Used SQL for missing values, outliers, and type casting, and Python for advanced EDA and statistical insights like correlation and hypothesis testing.
- **Data Transformation (SQL):**
  Extracted, transformed, and aggregated data, generating statistical summaries like averages and trends.
- **Visualization (Power BI):**
  Created dashboards showcasing inventory, procurement, and financial metrics, highlighting trends and distributions.
- **Presentation (PowerPoint):**
  Summarized findings with Power BI visuals and statistical insights for stakeholder communication.

# Project Architecture

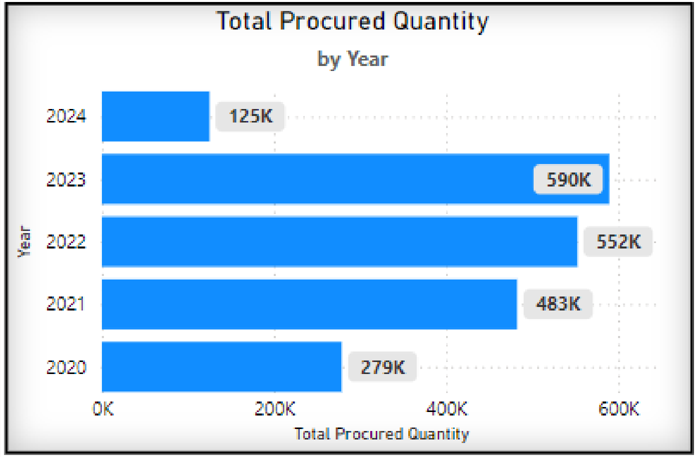


# Stock Levels Overview

- ❑ **Available Stock Metrics:**
- ▪ **Available Balance**: ₹357.92M
- ▪ **Available Quantity**: 211,000 units
- ▪ **Total Procured Quantity (2020-2024)**: 2,028,000 units
- ▪ **Key Insight**: Only **10.4%** of total procured stock is available, indicating high consumption or potential inefficiencies.

- ❑ **Yearly Consumption Trends (2020-2024):**
- ▪ **Peak Consumption**: 2022 and 2023 with **-523K** and **-522K** units, respectively.
- ▪ **Decline in 2024**: **-105K** units, a **-80%** decrease from 2023, suggesting improved stock utilization or a demand shift.

- ❑ **Yearly Procurement Trends (2020-2024):**
- ▪ **Peak Procurement**: 2023 with **590K** units, a **79%** drop in 2024 to **125K** units.



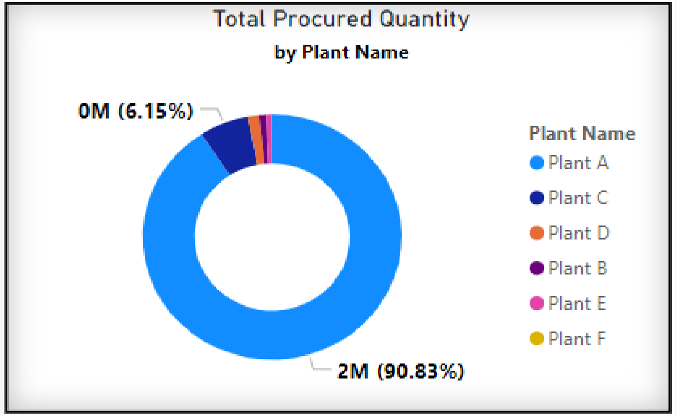
# Procurement Insights

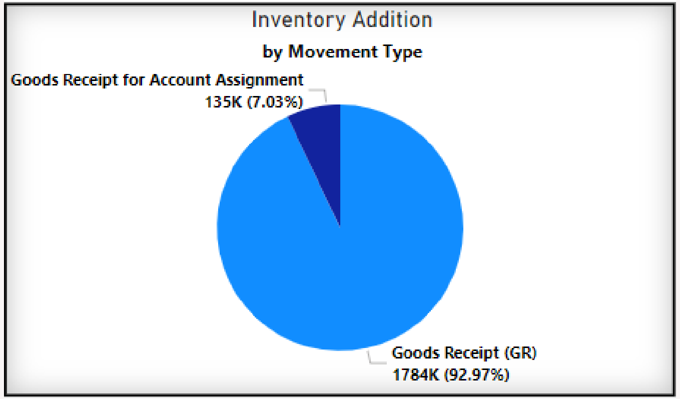
- ❑ **Procurement by Plant (2020-2024):**
- **Plant A**: Dominated procurement with **1.84M** units, accounting for **90.8%** of the total procurement.
- **Key Insight**: Plant A is the central hub for procurement, making it critical for focused inventory optimization and cost control measures.

- ❑ **Top Materials by Total Procurement:**
- **High-Speed Diesel**: **581,000** liters (28.7% of total procured stock).
- **Cable Ties (150mm)**: **85,000** units.
- **Chemical-Chlorination**: **74,000** kilograms.
- **Key Insight**: High-Speed Diesel is the most critical material, accounting for **28.7%** of total procurement, and should be monitored closely for optimization opportunities.

- ❑ **Procurement by Storage Location:**
- **Electrical Supplies**: **626,621** units (30.9% of total procurement).
- **Procurement Storage**: **182,147** units.
- **Road Delivery**: **179,094** units.
- **Key Insight**: Electrical Supplies represent a major portion of the procurement, indicating the need for a targeted approach in this category for better utilization.

- ❑ **Procurement Movement Type Breakdown:**
- **Goods Receipt (GR)**: **87.91%** of movements, totaling **1.78M** units.
- **Goods Receipt for Account Assignment**: **6.65%**.
- **Transfer Within Plant**: **5.12%**.
- **Key Insight**: The majority of procurement movements are Goods Receipts.



# Financial Insights

**❑ Cost Impact on Inventory (2020–2024)**

- **2020**: ₹756 million
- **2024**: ₹159 million (78.99% reduction)

  **Key Insight:** A significant reduction in inventory costs from 2020 to 2024, reflecting improved cost control and operational efficiencies.

**❑ Total Cost by Unit of Entry Category**

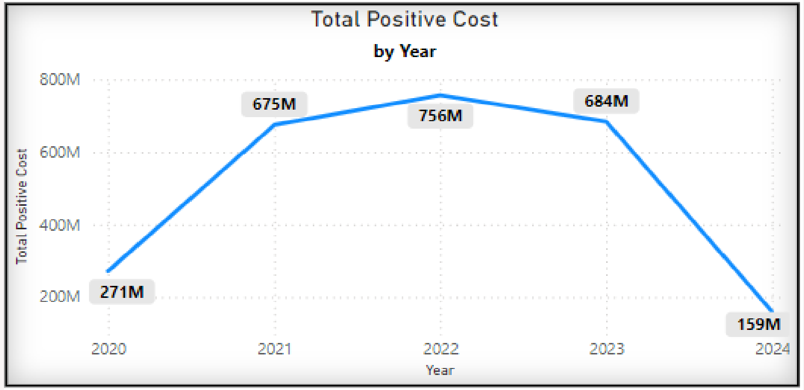
- **Quantity**: ₹60.4 million (42.83% of total costs)
- **Weight**: ₹56.3 million (40.76% of total costs)
- **Packaging**: ₹16.8 million (11.98% of total costs)
- **Measurement**: ₹6.2 million (4.43% of total costs)

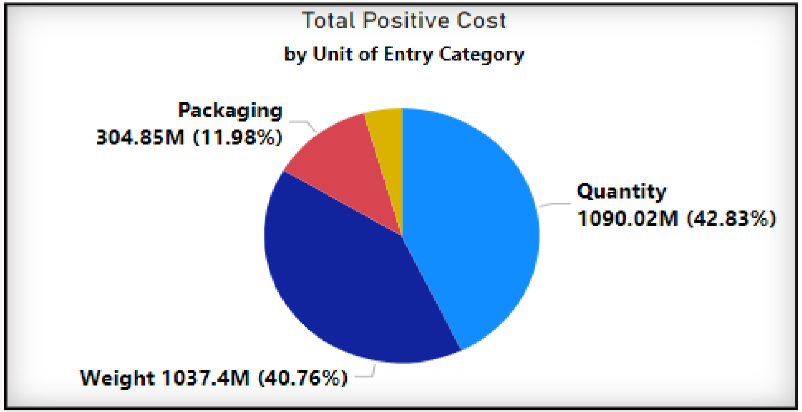
  **Key Insight**: **Quantity** and **Weight** contribute to the highest costs. Optimizing these categories can drive further savings.

**❑ Total Cost by Plant Name**

- **Plant A**: ₹781 million (88.54% of total costs)
- **Plant C**: ₹58 million (6.55% of total costs)
- **Other Plants (B, D, E, F)**: ₹42 million (5.91% of total costs)

  **Key Insight**: Most costs are concentrated in **Plant A** (88.54%). Cost optimization should focus on **Plant A** to achieve significant savings.



# Material Utilization and Efficiency

❑ **Top Materials by Total Procured Quantity**

- **High-Speed Diesel**: 581,000 liters (28.7% of total procured stock)
- **Cable Ties (150mm length)**: 85,000 units
- **Chemical-Chlorination**: 74,000 kilograms

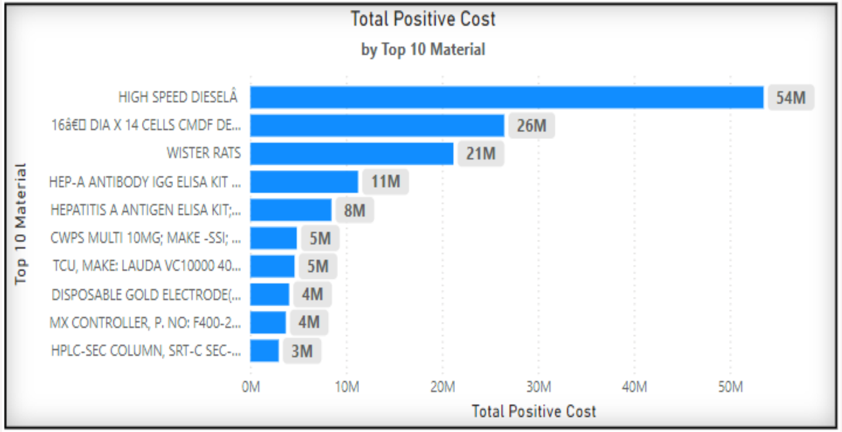
**Key Insight**: High-Speed Diesel is the most critical material, accounting for nearly 29% of total procurement. This material must be prioritized for cost and utilization optimization.

| Top 10 Material | Record Count | Total Procured Quantity | Units |
|---|---|---|---|
| HIGH SPEED DIESELÂ | 66 | 581000 | Liters |
| Cable Ties 150mm length | 17 | 85000 | Number |
| 1XDPBS WITHOUT CACL(GIBCO CAT#14190144)5 | 25 | 84500 | Milliliters |
| CHEMICAL-CHLORINATION ( INTECH ) | 192 | 74000 | Kilograms |
| ETHYLENE GLYCOL FOR -5 DEG. CHILLER | 54 | 63710 | Kilograms |
| CHEMICALS-SMBS(INTECH) | 196 | 30400 | Kilograms |
| MS CHEQUERED PLATE -5MTR*1.25MTR*5MM | 20 | 28310 | Kilograms |
| CHEMICALS-HCL COMMERCIAL GRADE | 81 | 23590 | Kilograms |
| TRYPLE (TM) SELECT INVITROGEN 125 | 6 | 20500 | Milliliters |
| FURNACE OIL | 2093 | 20396 | Kilotons |
| FURNACE OIL | 222 | 2021 | Tote |
| **Total** | **2972** | **1013427** | |

❑ **Top Material Cost Insights**

- **High-Speed Diesel**: ₹53.51M (38% of total costs)
- **Wister Rats**: ₹21.17M (15% of total costs)
- **16â€ DIA X 14 CELLS CMDF DEPTH CLEAR MODU**: ₹26.47M (18.8% of total costs)
- **HEP-A ANTIBODY IGG ELISA KIT**: ₹11.23M (8% of total costs)
- **HEPATITIS A ANTIGEN ELISA KIT**: ₹8.45M (6% of total costs)

**Key Insight**: **High-Speed Diesel** and **Wister Rats** are the top two contributors, representing a combined 53% of total material costs. Optimizing procurement and cost management for these materials can drive significant savings.



# Overstock and Understock Analysis

- **Overstock Issues:**
  Bois Storage is overstocked by 45.2%, leading to high storage costs.
  **Insight:** This overstocking costs **₹341.7 million annually**.
- **Understock Issues:**
  Plants F and D are understocked, with minimal inventory levels causing delays.
  **Insight:** This understocking adds **₹5.14 million annually** to procurement and production costs.
- **Inventory Imbalance:**
  45.2% of the inventory is overstocked, while 0.68% is understocked.
  **Insight:** These imbalances disrupt operations and lead to inefficiencies.
- **Cost Impact:**
  Overstocking costs **₹341.7 million per year**, while understocking adds **₹5.14 million** in unexpected expenses.
  **Insight:** Proper inventory management could save **₹346.8 million annually**.

# **Recommendations:** Strategic actions to achieve business objectives

- **Reduce Excess Inventory (Target: Reduce by 20%)**
  - **Action**: Focus on reducing stock in Electrical Supplies (30.9%) and High-Speed Diesel (28.7%).
  - **Impact**: This will free up space and reduce the costs of storing excess materials.
- **Improve Stock Usage (Target: Increase by 25%)**
  - **Action**: Optimize the usage of **High-Speed Diesel** (₹53.51M) and **Wister Rats** (₹21.17M), which together account for **53%** of the total material cost.
  - **Impact**: By using these materials efficiently, we can avoid wastage and save on purchasing new stock.
- **Align Procurement with Demand (Target: Reduce by 30%)**
  - **Action**: Purchase materials based on actual demand, especially for Plant A (90.8% of total procurement).
  - **Impact**: This will prevent over-purchasing and reduce unnecessary stock buildup.
- **Cut Costs at Key Plants (Target: Reduce by 15% at Plants A)**
  - **Action**: Focus cost optimization efforts on **Plant A** (88.54% of total costs).
  - **Impact**: Reducing costs at these plants will help save money and improve overall profitability.
- **Save on Procurement Costs (Target: Reduce by 20% in Quantity-based Materials)**
  - **Action**: Negotiate better deals for high-cost materials, like those in the Quantity category (76.99% of costs).
  - **Impact**: Reducing expenses at Plant A will yield significant savings, as it contributes the majority of costs.
- **Speed Up Material Turnover (Target: Improve by 15%)**
  - **Action**: Optimize turnover for **High-Speed Diesel** and **Wister Rats** (53% of total costs).
  - **Impact**: Faster turnover will reduce storage time and related costs.

By following these steps, we can reduce costs, use stock more effectively, and make procurement more efficient, directly helping to achieve the goal of cutting down on inventory costs and increasing stock usage.

# **Conclusion:** Summary of findings and next steps

❑ **Summary of Findings:**

- **Inventory Utilization:** Only 10.4% of total procured stock is available, indicating high consumption or inefficiency.
- **Procurement Insights:** Procurement is concentrated in Plant A (90.8%) and Electrical Supplies (30.9%), which should be optimized for better control.
- **Cost Control:** Inventory costs dropped by 87.9% from 2020 to 2024. However, Plant A and C account for 95.09% of costs, indicating significant opportunities for further improvement through targeted cost optimization strategies at these plants.
- **Material Turnover:** High-cost materials like Wister Rats and High-Speed Diesel make up a large portion of costs, presenting opportunities for cost reduction.

❑ **Next Steps:**

- **Optimize Inventory**: Reduce overstock and improve turnover rates to lower storage costs.
- **Align Procurement with Demand**: Focus on Plant A and Electrical Supplies to match procurement with actual usage.
- **Target Cost Reduction**: Work on reducing costs by 15% in Plants A and C.
- **Negotiate Better Material Deals**: Focus on reducing costs for high-cost materials like Wister Rats and High-Speed Diesel by 20%.

Focusing on these areas will help achieve objectives, reduce costs, and enhance stock utilization, ultimately improving overall efficiency.

# Thank You!
# For your time and consideration

Best Regards,
Jay Bourasi